Project Step 3 Draft Version: Design UI Interface + DML SQL (Group / On Ed Discussion)

- Due Feb 12 by 11:59pm
- Points 100
- Submitting a website url
- Available until Feb 13 at 11:59pm

This assignment was locked Feb 13 at 11:59pm.

Aim

In this step you create the UI front-end of a web app to provide all the CRUD functionalities as described in the **CS340 Project Guide (https://canvas.oregonstate.edu/courses/2031764/pages/cs340-project-guide)** while also writing the queries that you want to run on the Database for providing each functionality. To be clear we do not expect a fully functioning app, just the front-end UI pages for your entities and enough backend code to serve up those pages. You may also begin to implement SELECT queries for displaying the data from tables on your web pages, but we do not expect this to be complete. So basically, in this step, you are creating the pages of your website to support SELECT, INSERT UPDATE and DELETE. But with only enough back-end code to make these pages browsable. You will implement all other operations in later project steps.

Responsible use of AI tools is permitted on this assignment to help you write your Web Application. We suggest **Microsoft Copilot (https://copilot.microsoft.com/)**, an **OSU data protected tool. (https://canvas.oregonstate.edu/courses/1965953/pages/microsoft-copilot-for-the-web)** It will still be up to you to verify that the code is syntactically correct and returns the expected results. If you use one of the available tools to generate answers, you are required to say so in the submission and provide the prompt you used.

Deliverable

A **ZIP** archive containing the 4 THINGS described below in detail:

1. A **.PDF** file containing information from previous steps
2. A **URL** to a webpage (e.g. index.html) listing all your UI pages. Put this URL on the top of your PDF file as well as on your discussion post.
3. The **.SQL file containing your Data Manipulation Queries.** These are the queries that your website uses to let your users interact with data. Thus SELECT, INSERT, UPDATE and DELETE queries to provide the functionalities described in the **CS340 Project Guide (https://canvas.oregonstate.edu/courses/2031764/pages/cs340-project-guide)**.
4. The **.SQL file containing your Data Definition Queries** and the **sample data INSERT**

statements from the previous step

Make sure to name your zip file like **projectgroupX_stepX_DRAFT/FINAL.zip.** e.g. **ProjectGroup42_Step3_DRAFT.zip**

PDF File

a) Project Outline and Database Outline, ERD Schema & Sample Data Updated Version

This section should contain the updated outline, ERD, Schema and sample data based on the feedback from the grader and your peers as well as any design decisions that you decided to make on your own. It's required that you apply the various design tools you learned until now (like Normalization steps, **ON DELETE CASCADE (https://canvas.oregonstate.edu/courses/2031764/pages/exploration-mysql-cascade)** , etc.) to review and fix your ERD and Schema. Your Final Project submission will be graded based on your application of these concepts.

b) Fixes based on Feedback from Previous Steps:

This section in your PDF should detail what all things you were told to fix in the feedback by the grader and your peer reviewers in previous steps and how you fixed them. If you chose not to fix things based on the feedback, this section would contain the reasons why you think things stay as they are. If you didn't receive any feedback for a specific step from grader or reviewers, please mention that.

Apart from this, if you are making any changes based on your own changed design decisions, they should also be listed here along with the reasoning.

URL to a web start page

You need to **submit a URL of an index page linking all the other UI pages** in your website (e.g. http://classwork.engr.oregonstate.edu:9124) with a short but informative title for each page like "Browse existing ships", "Add new ship", "Manage certifications for each character".

- Your task this week is to translate your DB Schema into various UI pages to fulfill the CRUD requirements of the project and build a server app running with a port number on the ENGR server, similar to what you did in **Activity 2 - Connect webapp to database (https://canvas.oregonstate.edu/courses/2031764/assignments/10323319)** .
- To accomplish this you will use *the templating engine of your choice (e.g. .HBS handlebars templating engine) with a bare-bones Node/Python app. Otherwise, if you are using React, a frontend React server will suffice for meeting the minimum requirements of this step.*
- You will create this web app on the ENGR server and turn in an ENGR URL+port number for this project step.

- You only need to implement enough server-side code to make your pages browsable.
- If any of that sounds confusing, just ask on Ed Discussions or Teams so we can help!

These are the UI pages which the user will use to interact with your Project Website. Your UI pages should be the front-end implementation for all the functionalities described in the **CS340 Project Guide (https://canvas.oregonstate.edu/courses/2031764/pages/cs340-project-guide)**. These can be just basic pages built using the templating engine of the web technology (e.g. handlebars files) you choose. In later project steps you will add the server-side code to implement CRUD steps (e.g. SELECT INSERT, UPDATE and DELETE).

So basically, you need to create the client-side version of your site containing forms, any client-side validation that you want to perform (though not required), and any images, etc. that you want to be part of the website. Thus, the project website with minimal back-end code.

You can simply deploy this website on the ENGR server and create a server running with a port number as you did in **Activity 2 - Connect webapp to database (Individual) (https://canvas.oregonstate.edu/courses/2031764/assignments/10323319)**.

You need to include this URL in your PDF on the first page as well as on your discussion post.

For a sample website using the BSG database, I would turn in the URL to an **index.html** page listing all the other pages in the website with a short description of what functionality is available in what page.

One .SQL file should contain the Data Manipulation Queries:

These are the queries that your website uses to let your users interact with data. Thus SELECT, INSERT, UPDATE and DELETE queries to provide the functionalities described in the **CS340 Project Guide (https://canvas.oregonstate.edu/courses/2031764/pages/cs340-project-guide)**.

Anything that is a variable that you expect the user to fill or to be computed by your back-end code and passed to MySQL/MariaDB should be shown using some special characters around it. This variable is generally used by your back-end code to pass some data to SQL server. You can use any bunch of special characters. Just make a SQL comment (like in the example below) showing what characters you will be using to denote the sample part.

For example,the **bsg_sample_data_manipulation_queries.sql (https://canvas.oregonstate.edu/courses/2031764/files/115358915?wrap=1) (https://canvas.oregonstate.edu/courses/2031764/files/115358915/download?download_frd=1)**

contains the following query.

```
-- Query for add a new character functionality with colon : character being used to
-- denote the variables that will have data from the backend programming language

INSERT INTO bsg_people (fname, lname, homeworld, age)
```

```
VALUES (@fnameInput, @lnameInput, @homeworld_id_from_dropdown_Input, @ageInput);
```

Here, @fnameInput is a way to denote the variable that will have the value of First Name in it, when you later write the PL/SQL code to processes it. Again, you can choose any such way of denoting these variables.

We only want to see SQL here. Do *not* include any of the JavaScript or PHP or any other programming language's code used to process the data.

All queries should be syntactically correct (apart from the variable part described above) and relevant to other parts of your Project and the version of Data Definition Queries that you submit here. This means all your queries should work if the variable names are replaced with actual data.

How do I turn in this assignment?

1. Go to Ed Discussion and create a post called "Group XX Project Step 3 Draft". categorize your posting as *Project Step 3 Drafts & Reviews*. Communicate with your team so that you know who is posting the draft. Be sure that include *Both members' names and your Group number.*
    1. *Post the deliverables described (the ZIP file and the URL of your website) in the Discussions inside the Ed Discussion post for Project Step 3 Draft Version.*
2. Next, copy the URL of your Ed post and post it here on Canvas. Only one teammate needs to submit for the group. Failing to post to canvas will result in a deduction.
3. Each team member should then **review two other drafts (https://canvas.oregonstate.edu/courses/2031764/assignments/10323338)** -reviews are an important part of the process you are helping others while you are also learning. Strive to make thoughtful and substantial suggestions.

How should I get started?

I would suggest making a list of all CS340 Project functionalities in terms relevant to your database. For example "Add entries to each table" could be "Add a new BSG person", "Add a new Certification", and so on.

Then, draw a prototype of how you would your UI page look like. For example, an "Add new BSG person" page would contain fields for first name, last name, a dropdown for Homeworld, and a field for age. Next use the starter code for the templating engine of your choice and modify the example page to suit your schema entities.

Video

Frequently Asked Questions:

Q. ***I am using handlebars/some other templating engine for developing my project website. What should I submit?***

A. Just make sure the handlebar pages are browse-able with the right template syntax inserted.

For example, for the sample web app, if I were to submit only the handlebar file for bsg_people, I would make available the people.handlebars file with the URL I submit as well as an index.html file listing all the pages and a short description of what they are.

Q. ***How many web pages should my project have?***

A. There is neither a minimum or maximum number of UI pages that your project should have but it is desirable that your project is spread across different pages rather than a single one. You may combine or separate functionalities from the **CS340 Project Guide (https://canvas.oregonstate.edu/courses/2031764/pages/cs340-project-guide)** across pages as you deem necessary.

Q. ***Do I need to show sample data on the UI pages?***

A. Sample data from the previous project step would be great to illustrate how your pages will look and function. Alternatively, you could also begin to implement a SELECT query to populate your template page with the data from the table. It is not required to implement SELECT at this stage; just static sample data will be adequate.

Q. Do I need a navigation bar? What about page headings?

A. Each webpage should clearly indicate what entity is being displayed and each subsection of the

page should display what CRUD operation is going to occur. We are not too worried about how nicely all that looks as long as you have the CRUD elements on the page and each page is browsable.

Q. This is too much work; I can't implement my entire schema in a web app in one week!

A. This is a draft project step; and your app is a work in progress. You get credit for turning in something even if its incomplete. At this stage we do not expect a fully functioning web app, just one page for each entity and the forms for the CRUD operations you plan to implement. You will be finishing those up for in later project Steps.

Q. ***What do you mean by "front-end implementation for a functionality" on a UI page?***

A. For example, consider the Add people form for bsg_people entity. The UI page would contain a form with fields for all the columns (but not the ID since it's auto-incremented) and also a submit button. Of course, you would also have labels telling what each of these textboxes or dropdown are for. And a header on the page specifying that this form is for adding new person. You could also go one step ahead and make sure that the fields are displayed in a neat order, probably using a table since there will be points for styling in your Final Project submission.

Q. ***So these UI pages will interact with the database?***

A. Well, they will actually interact with your backend programming language or framework like Flask/NodeJS which then interacts with the database. More on that in Week 7. We will get there soon enough.

Q. ***Should this particular field in my form be a textbox/textarea/drop-down/some new UX style fancy element ?***

A. This is your choice as the user should easily be able to understand how to use it (maybe if it's too difficult you could provide tooltips about how to use it). Your reviewers/graders might disagree and you should be able to justify the usage of one over the other.

Q. ***How "done" does the design of the UI pages need to be for this step?***

A. "Done" could be answered by asking the question: Does the page provide all the necessary UI elements for implementing a specific functionality? For example, the BSG People page would provide all the things necessary for implementing the Add New row to the bsg_people.

Q. ***My website front-end uses AJAX to display rows and the website can't function without it/ I have already written the server-side code to make my website work. Do I need to remove/ hide these functionalities?***

A. You do not have to remove/hide any "extra" functionality or features for this submission that makes your front end work. You also do not need to scale down your project or disconnect the server-side code if it's already working. It's perfectly fine to implement a website with more

functionalities than required for this Step.

All those extra things will simply not be considered by your reviewers and the graders when the Final version is graded. For example, the form's fields, the presence of a Submit button and the presence of relevant headers and labels on your form to make sure that anyone looking at the form understands what it is for and how to interact with the form, will be the only things reviewed and graded.

Q. ***Why do we have to work on this UI web page thing? Isn't this course about databases? Why not just let me write SQL and get points for it?***

A. Databases do not exist in isolation. Unless all your users are comfortable writing SQL for tasks like inserting, and browsing data, you definitely need to provide them with an interface to interact with this database. And that is where web applications come into the pictures, as described in **Exploration - Three Tier Web Architecture (https://canvas.oregonstate.edu/courses/2031764/pages/exploration-three-tier-web-architecture)** . Forms facilitate a way for providing data to put in your INSERT and UPDATE statements. The data that you get using SELECT statements will be displayed using tables in your UI pages. And there probably will be a button somewhere which allows a user to DELETE a record in your table.

Q. ***I have already learned database interaction using a web technology like NodeJS in CS290. Why am I learning this now?***

A.CS290 offers an introduction to database interaction using a web technology, while this course gives you an opportunity to create a full-fledged database-drive website.

Q. ***My question was not answered in this Q&A. Where can I go to find the answer?***

A. Ask on Ed Discussion . We update the FAQ sections every quarter.

Points

This is worth 100 points.

You get points for simply turning in the work.

Step 3 Draft

Criteria	Ratings				Pts
Project Draft ZIP	**30 pts** **Full Marks** Includes ALL required elements 1) zip file correctly named, 2) PDF with information from previous steps 3) DML.SQL file with queries, and 4) DDL.SQL file with sample data	**24 pts** **Missing ONE** Missing no more than ONE required element 1) zip file correctly named, 2) PDF with information from previous steps 3) DML.SQL file with queries, and 4) DDL.SQL file with sample data	**18 pts** **Missing TWO** Missing no more than TWO required element 1) zip file correctly named, 2) PDF with information from previous steps 3) DML.SQL file with queries, and 4) DDL.SQL file with sample data	**12 pts** **Missing THREE or More** Missing THREE or more required elements 1) zip file correctly named, 2) PDF with information from previous steps 3) DML.SQL file with queries, and 4) DDL.SQL file with sample data	30 pts
Posts to discussion	**20 pts** **Full Marks** 1) Posts draft as a Group XX Project Step 3 Draft" 2) categorizes as Project Step 3 Drafts & Reviews 3) Includes both members' names and group number 4) has a URL to all UI pages.	**16 pts** **Missing ONE** Missing AT MOST ONE of 1) Posts draft as a Group XX Project Step 3 Draft" 2) categorizes as Project Step 3 Drafts & Reviews 3) Includes both members' names and group number 4) has a URL to all UI pages.	**12 pts** **Missing TWO** Missing AT MOST TWO of 1) Posts draft as a Group XX Project Step 3 Draft" 2) categorizes as Project Step 3 Drafts & Reviews 3) Includes both members' names and group number 4) has a URL to all UI pages.	**8 pts** **Missing THREE or More** Missing THREE OR MORE of 1) Posts draft as a Group XX Project Step 3 Draft" 2) categorizes as Project Step 3 Drafts & Reviews 3) Includes both members' names and group number 4) has a URL to all UI pages.	20 pts

Criteria	Ratings				Pts
UI web pages	**40 pts** **Full Marks** URL goes to UI web pages which the user will use to interact with your Project Website. EACH TABLE in schema has a corresponding page.	**36 pts** **Missing ONE** URL goes to UI web pages which the user will use to interact with your Project Website. AT MOST ONE table in schema does not have corresponding HTML page.	**32 pts** **Missing TWO** URL goes to UI web pages which the user will use to interact with your Project Website. AT MOST TWO tables in schema do not have a corresponding HTML page.	**28 pts** **Missing THREE or More** URL goes to UI web pages which the user will use to interact with your Project Website. THREE OR MORE tables in schema do not have a corresponding HTML page.	40 pts
Posts URL to Canvas	**10 pts** **Full Marks** Completes discussion posting by due date and Includes URL to discussion posting in Canvas submission.	**6 pts** **URL LATE < 24 hrs** Completes discussion posting by due date BUT ADDS canvas posting of URL to discussion posting in Canvas submission LATE < 24 hrs	**4 pts** **Canvas CMT w/ URL before graded** Completes discussion posting by due date BUT does not post to canvas. Adds canvas COMMENT with URL to discussion posting in Canvas submission before it has been graded.	**0 pts** **No Marks** Completes discussion posting by due date BUT does not post to canvas. Emails instructor/TA and adds canvas COMMENT with URL to discussion posting in Canvas submission AFTER it has been graded.	10 pts
					Total Points: 100